

Curriculum Vitae of



Mohammad Robiul Islam (M. Robiul Islam), MS Integrated PhD. Researcher Fellow

Date of Birth: 07 Oct 1997; **Gender:** Male;

Nationality: Bangladeshi

PhD Researcher Fellow in Electronic Engineering,
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<https://www.webofscience.com/wos/author/record/KLY-78012024>



Research Areas/Interests

- Triboelectric Nanogenerators, Self-powered sensor, Electromagnetic Energy Harvesting, Piezoelectric Nanogenerators, Biomechanical Energy Harvesting.
- Flexible, stretchable, breathable, and wearable bioelectronics
- 2D nanomaterials synthesis and functionalization (MXene, MOF, COF and graphitic carbon)
- Flexible electro-spun polymer nanofibers
- Tissue like soft hydrogel
- Wearable digital healthcare and Physical AI industries

Education Background

- **Doctor of Philosophy in Electronic Engineering**
2022-Present (Thesis writing)
Kwangwoon University, Seoul, Republic of Korea
Advanced Sensor and Energy Research (ASER) Laboratory
Supervisor: **Prof. Jae Yeong Park**
CGPA: 4.42 (out of 4.50)

- **Bachelor of Science in Electrical and Electronic Engineering**

March 2022

Islamic University, Kushtia, Bangladesh

CGPA: 3.48 (out of 4)

Position: 6th /50

- **Higher Secondary Certificate (HSC)**

May 2016

Savar Cantonment Public School & College, Dhaka

Dhaka Education Board, Bangladesh

CGPA: 5.00 (out of 5)

RESEARCH EXPERIENCE

- **PhD Researcher Fellow**

2022-Present (Thesis writing)

Advanced Sensor and Energy Research (ASER) Laboratory,

Dept. of Electronic Engineering, Kwangwoon University

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PUBLICATIONS (13)

1. **M. Robiul Islam**, Omar Faruk, S M Sohel Rana, Gagan Bahadur Pradhan, Hongseok Kim, Md. Selim Reza, Trilochan Bhatta, Jae Yeong Park, Poly-DADMAC Functionalized Polyethylene Oxide Composite Nanofibrous Mat as Highly Positive Material for Triboelectric Nanogenerators and Self-Powered Pressure Sensors, **Advanced Functional Materials**, Volume 34, Issue 40 2403899 (2024), [IF: 18.5, Q1]. DOI: <https://doi.org/10.1002/adfm.202403899>
2. **M. Robiul Islam**, Omar Faruk, Md Selim Reza, Trilochan Bhatta, Md Shofiul Alam, Zahidul Islam, SeungJae Lim, Jae Yeong Park, Amino-rich graphitic carbon nitride functionalized polyurethane nanomesh as highly tribopositive material for high-performance triboelectric nanogenerators and self-powered respiration monitoring wearables, **Chemical Engineering Journal** 518 (2025) 164564 [IF: 13.4, Q1]. DOI: <https://doi.org/10.1016/j.cej.2025.164564>
3. **M. Robiul Islam**, Hongseok Kim, Shital Sharma, Aklesh Teli, Kumar Shrestha, Omar Faruk, Moon Seong Jo, Trilochan Bhatta, Jae Yeong Park, Covalent Organic Framework-Functionalized Polyurethane-Based High-Performance Stretchable and Breathable All-Nanofibrous Triboelectric Nanogenerator for Wearable Sign Language Interpretation,

Advanced Energy Materials, 2025, e05692. [IF: 26.0, Q1].
DOI: <https://doi.org/10.1002/aenm.202505692>

4. Omar Faruk, **M. Robiul Islam**, S M Sohel Rana, Gagan Bahadur Pradhan, HongSeok Kim, Md Asaduzzaman, Trilochan Bhatta, Jae Yeong Park, V2CTX-MXene-functionalized fluoropolymer composite nanofibrous mat-based high-performance triboelectric nanogenerator for self-powered human activity and posture monitoring, **Nano Energy** 127 (2024) 109787(2024), [IF: 16.8, Q1]. DOI: <https://doi.org/10.1016/j.nanoen.2024.109787>
5. Omar Faruk, **M. Robiul Islam**, Gagan Bahadur Pradhan, Trilochan Bhatta, Md Asaduzzaman, Md Shofiul Alam, SeungJae Lim , Jae Yeong Park, A Wearable Triboelectric-Iontronic Hybrid Smart Finger Ring with Self-Powered Static-Dynamic Tactile Sensing for Advanced Human-Machine Interactions, **Advanced Functional materials**, 2025; 0:e27394, [IF: 19, Q1]. DOI: <https://doi.org/10.1002/adfm.202527394>
6. HongSeok Kim,S M Sohel Rana, **M. Robiul Islam**, Omar Faruk, Kumar Shrestha, Gagan Bahadur Pradhan, Hongseok Kim, Jae Yeong Park, A molybdenum-disulfide nanocomposite film-based stretchable triboelectric nanogenerator for wearable biomechanical energy harvesting and self-powered human motion monitoring, **Chemical Engineering Journal** 491 (2024) 151980, [IF: 13.4, Q1]. DOI: <https://doi.org/10.1016/j.cej.2024.151980>
7. Md Abu Zahed, S M Sohel Rana, Omar Faruk, **M. Robiul Islam**, Md Selim Reza, YeYeong Lee, Gagan Bahadur Pradhan, Md Asaduzzaman, Dongyun Kim, Trilochan Bhatta, and Jae Yeong Park, Self-Powered Wireless System for Monitoring Sweat Electrolytes in Personalized Healthcare Wearables, **Advanced Functional Materials**, 2024, 2421021, [IF: 18.5, Q1].DOI: <https://doi.org/10.1002/adfm.202421021>
- 8.
9. Trilochan Bhatta, Omar Faruk, **M. Robiul Islam**, Hong Seok Kim, SM Sohel Rana, Gagan Bahadur Pradhan, Akash Deo, Dae-Sung Kwon, Ilseon Yoo, Jae Yeong Park, Polymeric multilayered planar spring-based hybrid nanogenerator integrated with a self-powered vibration sensor for automotive vehicles IoT applications, **Nano Energy** 127 (2024) 109793, [IF: 16.8, Q1]. DOI: <https://doi.org/10.1016/j.nanoen.2024.109793>
10. Md Selim Reza ,Md Sharifuzzaman , Zahidul Islam , Md Assaduzaman , YeYoung Lee , Dongyun Kim , **M. Robiul Islam** , Hyeong Seok Kang , HongSeok Kim , Dae Heum Kim , Hyun-Do Jung , Jae Yeong Park , A flexible and multimodal biosensing patch integrated with microfluidics for chronic wound monitoring, **Chemical Engineering Journal** 501 (2024) 157673, [IF: 13.4, Q1]. DOI: <https://doi.org/10.1016/j.cej.2024.157673>

11. S M Sohel Rana, Omar Faruk, Md Selim Reza , **M. Robiul Islam**, Hongseok Kim, Jae Yeong Park, All porous Ecoflex and SEBS-based stretchable high-performance triboelectric nanogenerator for self-powered human activity monitoring, **Chemical Engineering Journal**, 488 (2024) 151050, [IF: 13.4, Q1]. <https://doi.org/10.1016/j.cej.2024.151050>
12. S M Sohel Rana; Md Abu Zahed; **M. Robiul Islam**; Omar Faruk; Hye Su Song; Seong Hoon Jeong; Jae Yeong Park, Metal-organic framework and molybdenum oxide doped highly negative hybridized triboelectric material for self-powered and continuous monitoring of biosignals, **Chemical Engineering Journal** 473 (2023) 144989, [IF: 13.4, Q1]. <https://doi.org/10.1016/j.cej.2023.144989>
13. S.M. Sohel Rana; Omar Faruk; **M. Robiul Islam**; Tamanna Yasmin; K. Zaman; Zhong Lin Wang, Recent advances in metal-organic framework-based self-powered sensors: A promising energy harvesting technology, **Coordination Chemistry Reviews** 507 (2024) 215741, [IF: 23.5, Q1]. <https://doi.org/10.1016/j.ccr.2024.215741>
14. Islam, Z.; Reza, M.S.; Asaduzzaman, **M. Robiul Islam**, M.R.; Samad, A.A.; Kim, D.; Alam, M.S.; Park, J.Y., ZIF-67/organometallic-derived Co/NPC@TiO₂ nanohybrids by laser pyrolysis for flexible skin-interfaced sweat-analysis multimodal patch, **Chemical Engineering Journal**, 522 (2025) 167121, [IF: 13.4, Q1]. <https://doi.org/10.1016/j.cej.2025.167121>

Conference Presentation (4)

1. **M. Robiul Islam**, Md. Shofiul Alam, and Jae Yeong Park, “All-Nanofibrous Breathable and Stretchable Triboelectric Nanogenerators for Sign Language Recognition,” 8th International Conference on Nanogenerator and Piezotronics (**NGPT 2026**), Sungkyunkwan University, Seoul, Republic of Korea, June 2026. (**Oral Presentation**)
2. **M. Robiul Islam**, Omar Faruk, Md Selim Reza, Trilochan Bhatta, Md Shofiul Alam, Zahidul Islam, SeungJae Lim, Jae Yeong Park, Melamine-Derived Carbon Nitride-Functionalized Polyurethane Nanomesh-Based Triboelectric Nanogenerator for Self-Powered Respiration Monitoring, 2025 23rd International Conference on Solid-State Sensors, Actuators and Microsystems (**Transducers**), Orlando, FL, USA, **Oral presentation**)
3. **M. Robiul Islam**, Omar Faruk, S M Sohel Rana, Gagan Bahadur Pradhan, and Jae Yeong Park, Cationic Polymer Functionalized Nanofiber Mat-Based Triboelectric Nanogenerator for Self-Powered Human Motion Monitoring, 2023 IEEE 22nd International Conference on Micro and Nanotechnology for Power Generation and Energy Conversion Applications (**PowerMEMS**), Abu Dhabi, UAE. (**Oral presentation**)

4. SM Soheli Rana, **M. Robiul Islam**, HongSeok Kim and Jae Yeong Park, Nanoporous cobalt oxide-based bifunctional triboelectric nanogenerators for self-powered pressure sensor, The 18th International Conference on Nano/Micro Engineered and Molecular Systems IEEE NEMS 2023, Jeju, Korea. (Oral presentation)

Conference Paper (6)

1. **M. Robiul Islam**, Omar Faruk, S M Soheli Rana, Gagan Bahadur Pradhan, and Jae Yeong Park, Cationic Polymer Functionalized Nanofiber Mat-Based Triboelectric Nanogenerator for Self-Powered Human Motion Monitoring, 2023 IEEE 22nd International Conference on Micro and Nanotechnology for Power Generation and Energy Conversion Applications (PowerMEMS), Abu Dhabi, UAE. DOI:[10.1109/PowerMEMS59329.2023](https://doi.org/10.1109/PowerMEMS59329.2023). (Oral presentation)
2. SM Soheli Rana, **M. Robiul Islam**, HongSeok Kim and Jae Yeong Park, Nanoporous cobalt oxide-based bifunctional triboelectric nanogenerators for self-powered pressure sensor, The 18th International Conference on Nano/Micro Engineered and Molecular Systems IEEE NEMS 2023, Jeju, Korea. (Oral presentation)
3. **M. Robiul Islam**, Omar Faruk, Md Selim Reza, Trilochan Bhatta, Md Shofiul Alam, Zahidul Islam, SeungJae Lim, Jae Yeong Park, Melamine-Derived Carbon Nitride-Functionalized Polyurethane Nanomesh-Based Triboelectric Nanogenerator for Self-Powered Respiration Monitoring, 2025 23rd International Conference on Solid-State Sensors, Actuators and Microsystems (Transducers) , Orlando, FL, USA ,DOI: [10.1109/Transducers61432.2025.11110261](https://doi.org/10.1109/Transducers61432.2025.11110261). (Oral presentation)
4. Omar Faruk, M Robiul Islam, SM Soheli Rana, Kumar Shrestha, Jae Yeong Park, V2CTx /PVDF-HFP Composite Nanofibers-Based Self-Powered Pressure Sensor for Human Foot Pressure and Activity Monitoring, 2023 IEEE 22nd International Conference on Micro and Nanotechnology for Power Generation and Energy Conversion Applications (PowerMEMS), Abu Dhabi, UAE. DOI: [10.1109/PowerMEMS59329.2023.10417521](https://doi.org/10.1109/PowerMEMS59329.2023.10417521)
5. Omar Faruk, Md Asaduzzaman, M Robiul Islam, Gagan Bahadur Pradhan, Md Shofiul Alam, Jae Yeong Park, Self-Powered Smart Finger Ring Based on Triboelectric-Iontronic Hybrid Pressure Sensor for Human-Machine Interfaces, 2025 23rd International Conference on Solid-State Sensors, Actuators and Microsystems (Transducers) , Orlando, FL, USA, DOI: [10.1109/Transducers61432.2025.11109162](https://doi.org/10.1109/Transducers61432.2025.11109162)
6. Md Selim Reza, Zahidul Islam, Md Asaduzzaman, Ahmad Abdus Samad, Dongyun Kim, M Robiul Islam, Jae Yeong Park, 3D-Flower-Like Bimetallic MXene Incorporated Laser-Engraved Graphene-Based Physio-Chemical Hybrid Patch for Perspiration Analysis and

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Java, Python
- **Simulation Language:** Matlab, Comsol physics
- **Microcontroller and Embedded Systems:** Arduino, PSpice
- **Graphic Design:** Adobe Photoshop, Auto CAD, Rhino, Materials studio
- **Office Work:** Microsoft Word, Microsoft Excel, Microsoft PowerPoint, and origin

GRANTS and AWARDS

- Recipient of **“Islamic University Merit Scholarship”** for two years during B.Sc. at the Islamic University, Kushtia-7003, Bangladesh.
- Finalist for the Outstanding Young Researcher Paper Competition at the 23rd International Conference on Solid-State Sensors, Actuators and Microsystems (Transducers’25) Conference

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.



(Mohammad Robiul Islam)

REFERENCES

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Published Cover Image (1)

